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12 – GGR – M - PG01 - Environmental Change and Natural Risks - A Biogeographical Perspective

6 - Natural risks

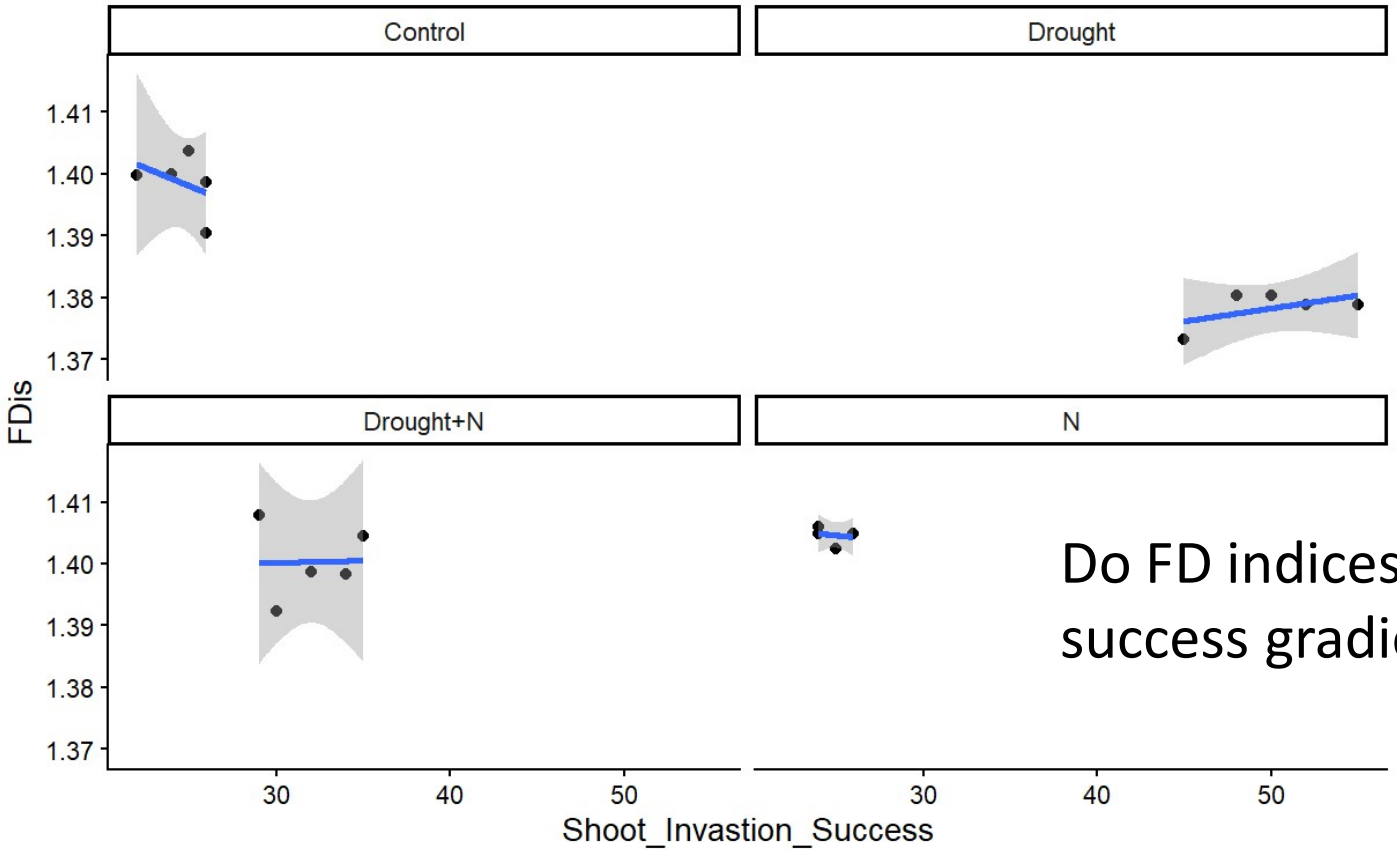
Leipzig, 19.01.2026

Claudia Guimarães-Steinicke
E-Mail: claudia.steinicke@uni-leipzig.de

Schedule

Dates	Content
21.10.2025	Introduction and seminar structure
3.11.2025	Multivariate analysis applied to tropical plant communities - Study case: Atlantic Rain Forest **(2,5 points)**
17.11.2025	Species distribution modeling (SDM) – Part 1
8.12.2025	Species distribution modeling (SDM) – Part 2 **(5 points)**
12.01.2026	Invasive species **(2,5 points)**
19.01.2026	**Natural risks and ecosystem stability**
26.01.2026	Biodiveristy loss and land use change (Q&A Session)

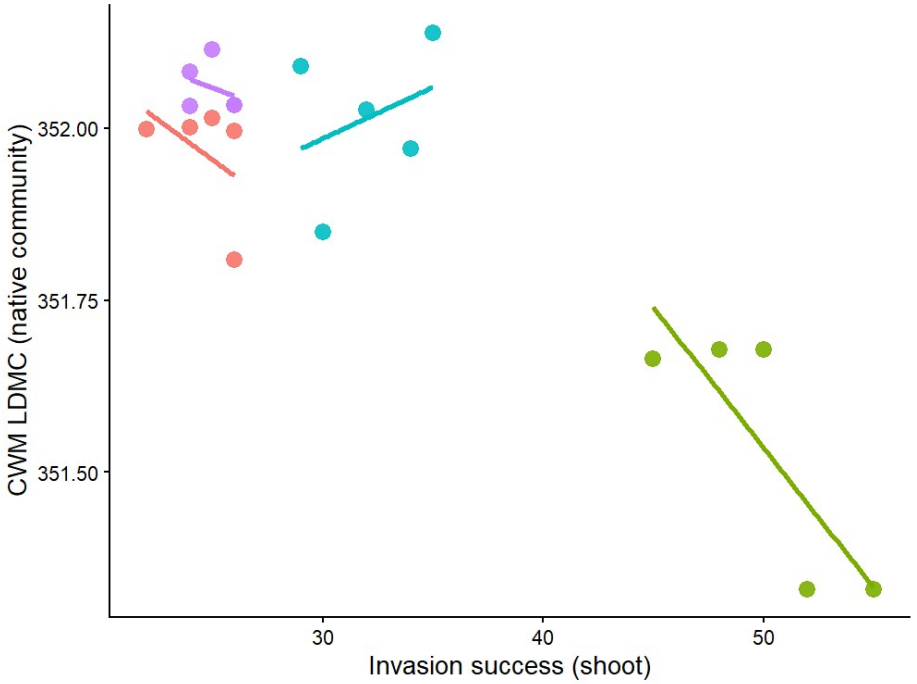
FEEDBACK FROM THE INVASIVE AND FD EXERCISES:

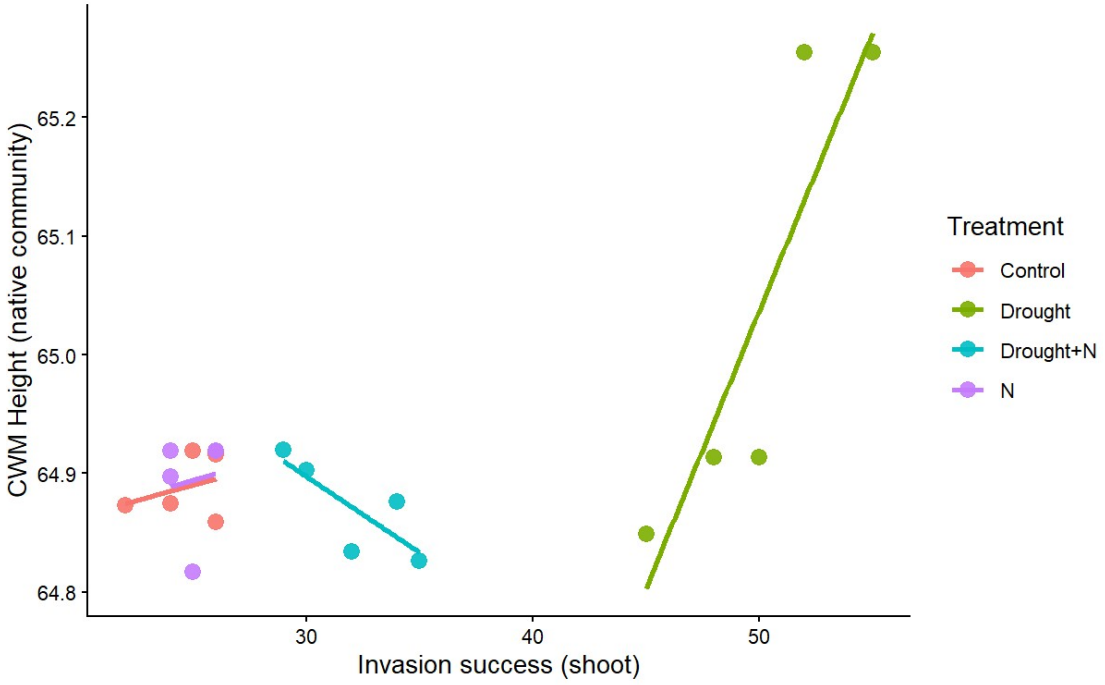


Do FD indices change along the invasion success gradient?

FEEDBACK FROM THE INVASIVE AND FD EXERCISES:

Do CWM traits change along the invasion gradient?

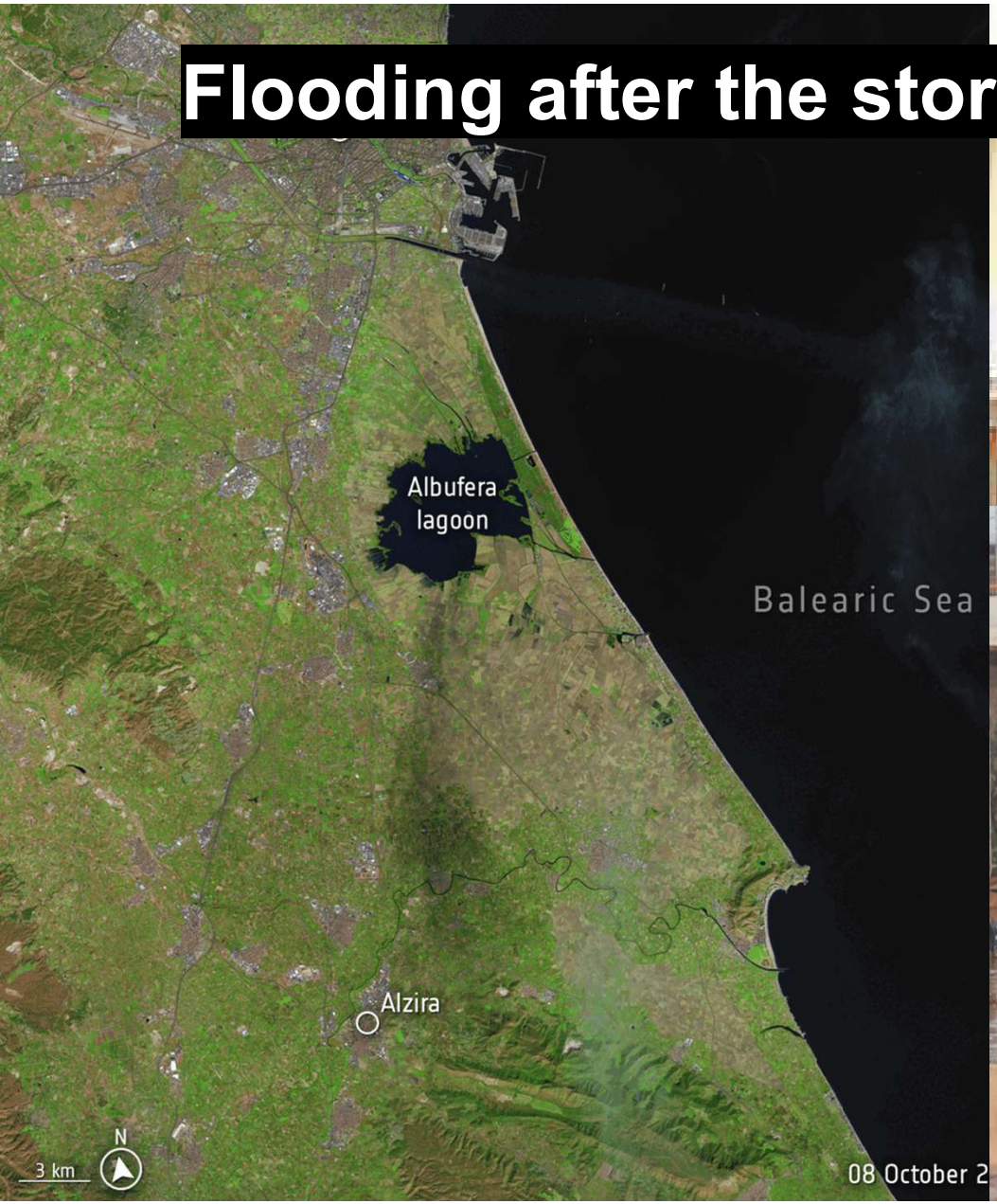




The goals of this seminar are:

- Understand the differences between natural hazards, disasters, and risks
- Overview of global data
- How do natural hazards are studied from a biogeography perspective?
- Ecosystem stability
- Properties and measures of ecosystem stability
- Working in groups – presentation about natural hazard events.

Flooding after the storm Briton in Valencia, 2024









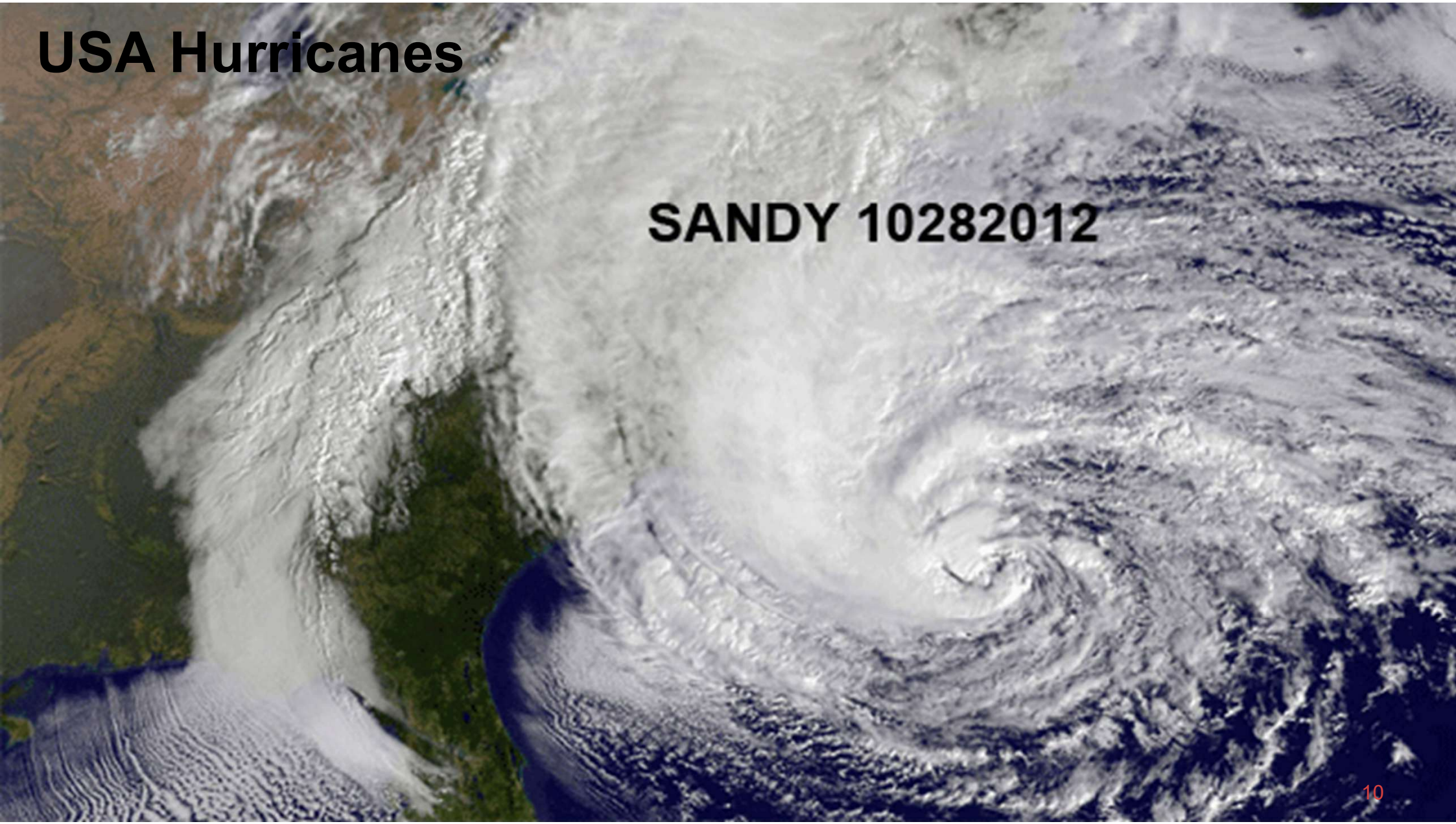
Flooding after THE storm in Ahrweiler, Germany, 2021


Earthquake in Turkey

Landslide in Brazil



USA Hurricanes

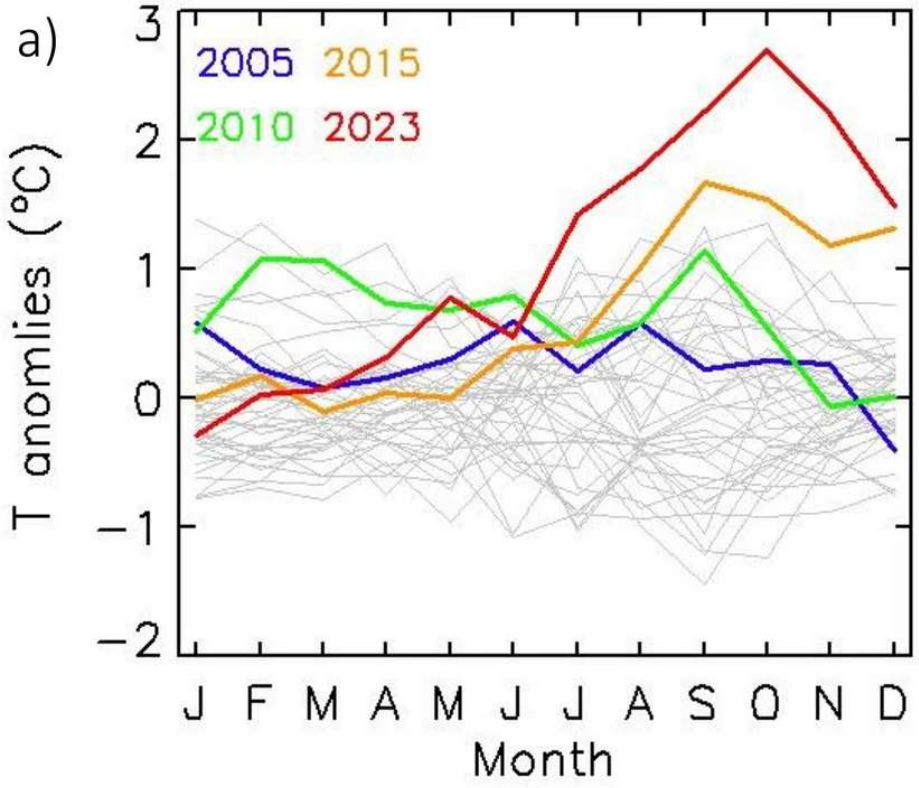


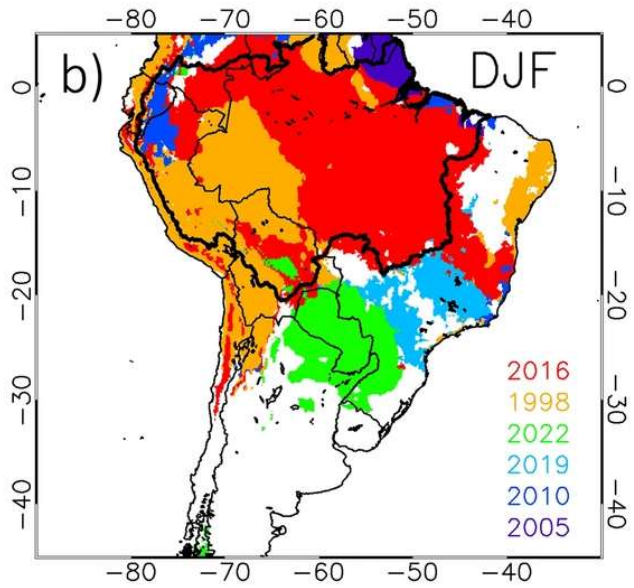

Wildfire Greece

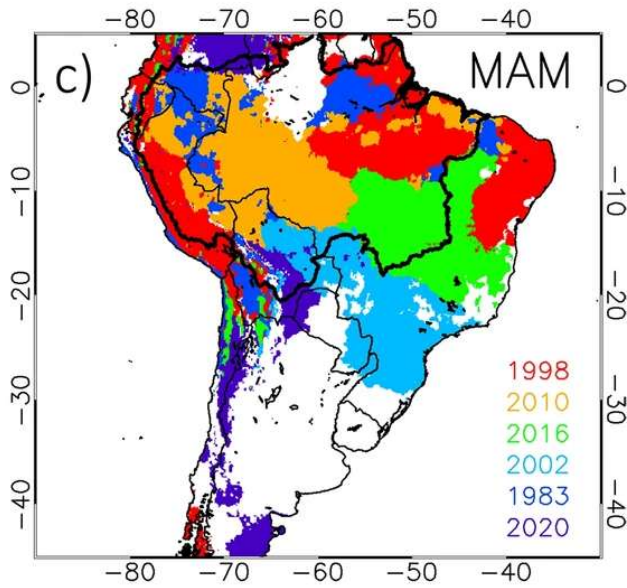

California

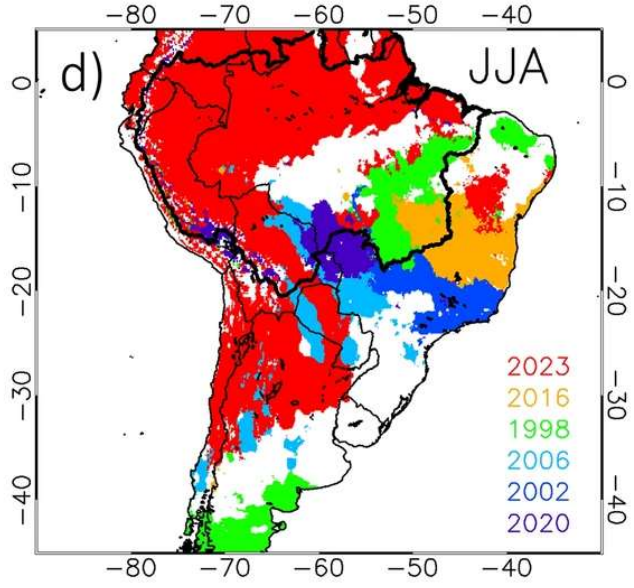

Portugal

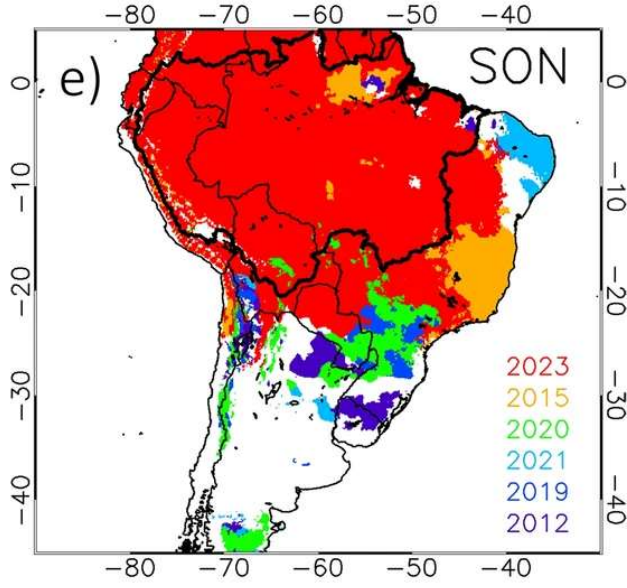

Wetland Brazil



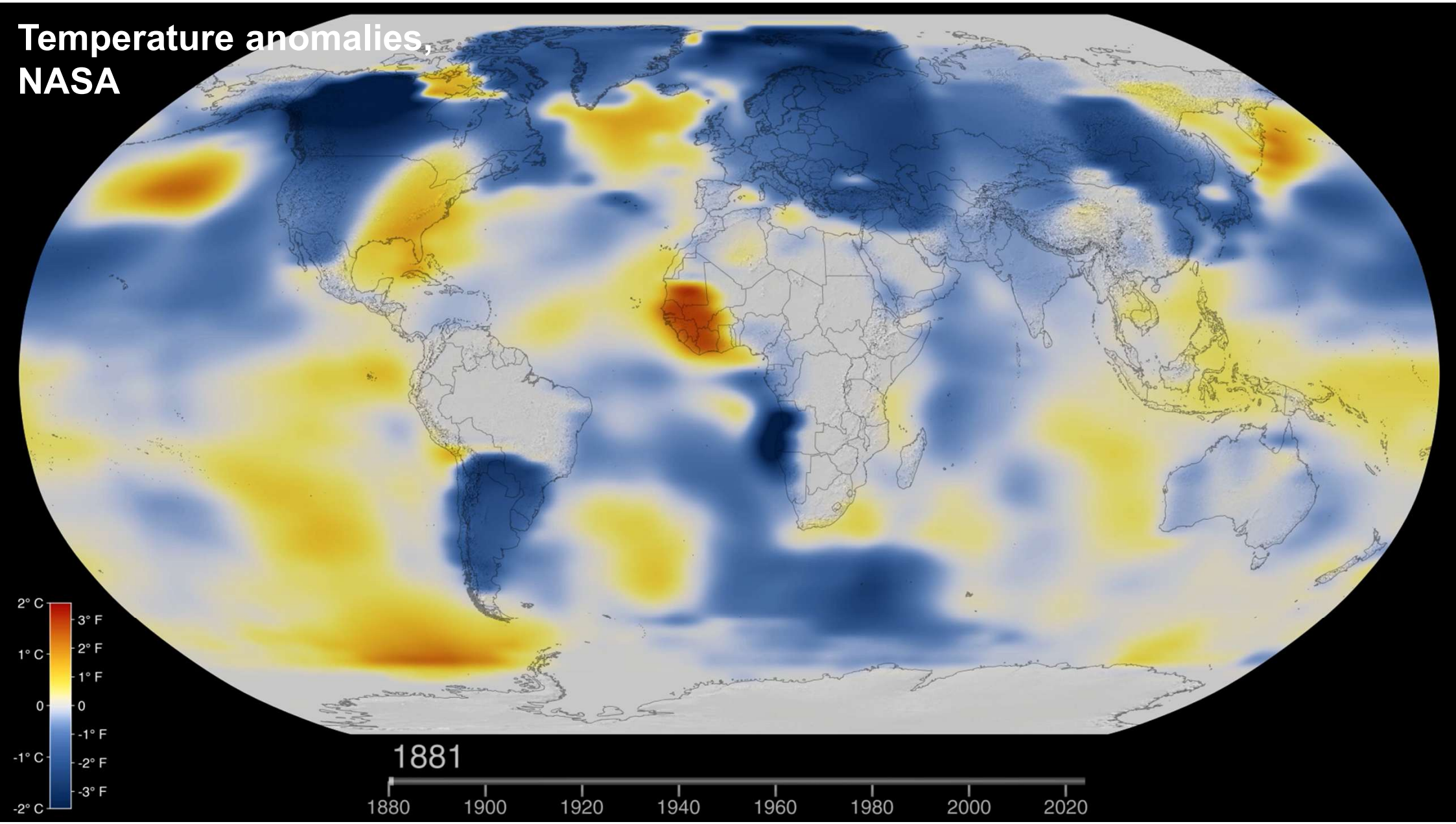








Espinoza et al. 2024 Scientific reports

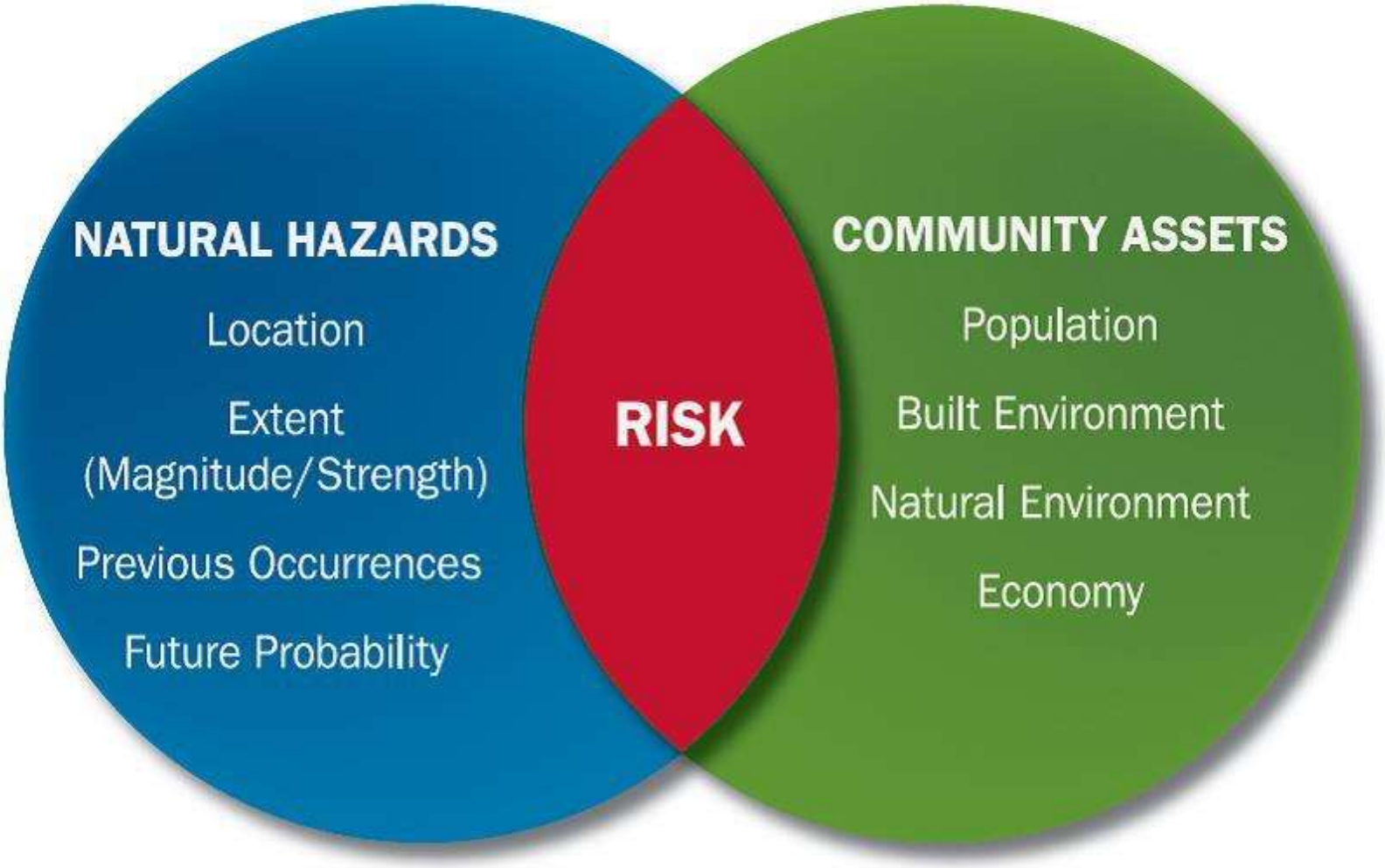

Temperature anomalies,
NASA
2° C
1° C
0
-1° C
-2° C
3° F
2° F
1° F
0
-1° F
-2° F
-3° F
1881
1880
1900
1920
1940
1960
1980
2000
2020

NATURAL HAZARDS, DISASTERS, AND RISKS

Concepts following UNISDR:

- **Hazards**: "a natural process or phenomenon that may negatively impact the economy, society, and ecology." Hazards are the origin of disasters.
  - Type of hazards by causes: Geophysical, hydrological, meteorological, biological…
- **Disasters**: "Direct or indirect results of hazards. Disaster impacts include human losses, property losses, resources and environmental destruction, ecological damages, and disruption of social order."
- **Risks: "**As the probability of harmful consequences resulting from interactions between natural or human-induced hazards and vulnerable conditions."

UNISDR 2014; Shi 2019, *Disaster Risk Science*.

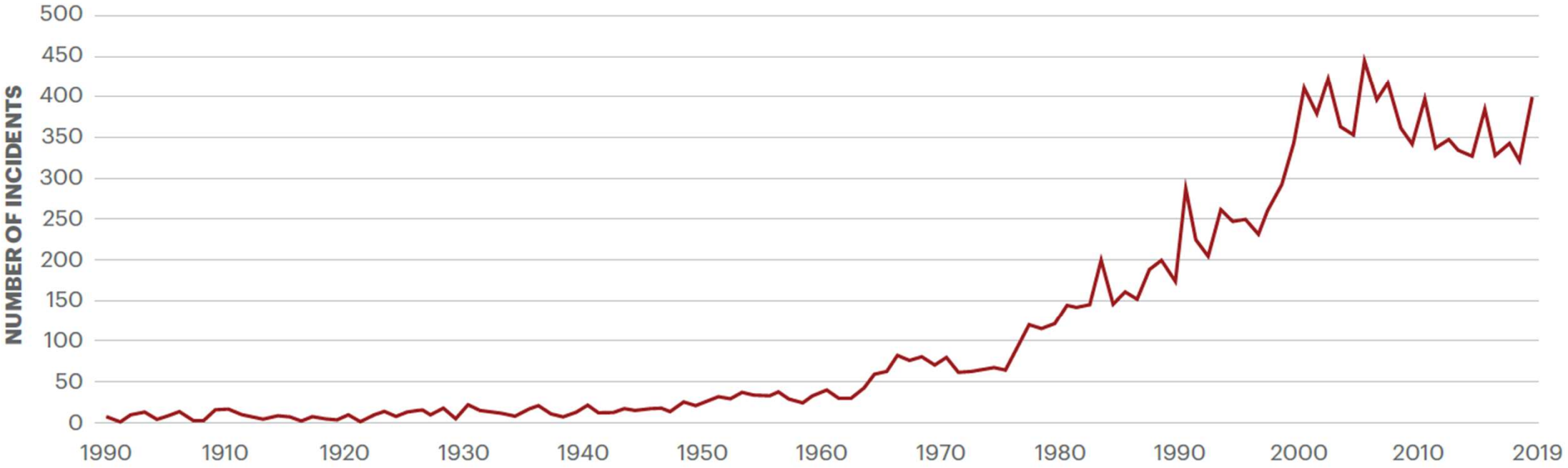


Source: www.planningforhazards.com

FIGURE 3.1

Trend in the number of natural disasters, 1900 to 2019

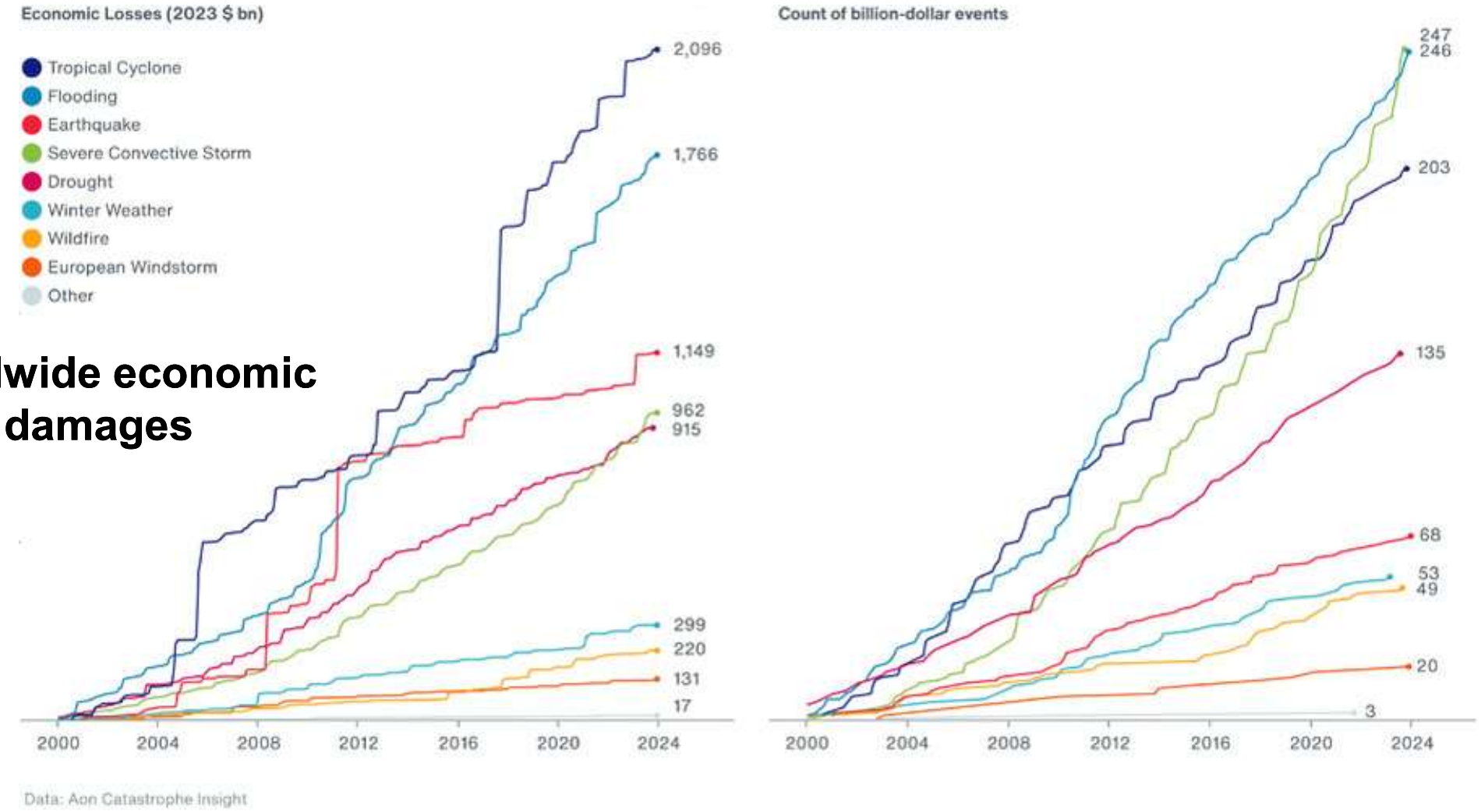
The global number of natural disasters has increased tenfold since 1960.



Source: EM-DAT
Note: Includes Drought, Earthquakes, Volcanic activity, Mass movement (dry), Storms, Floods, Landslides, Wildfire and Extreme temperature.

EM-DAT: The International Disaster Database, Centre for Research on the Epidemiology of disaster (University Louvain, Belgium)

Worldwide economic damages



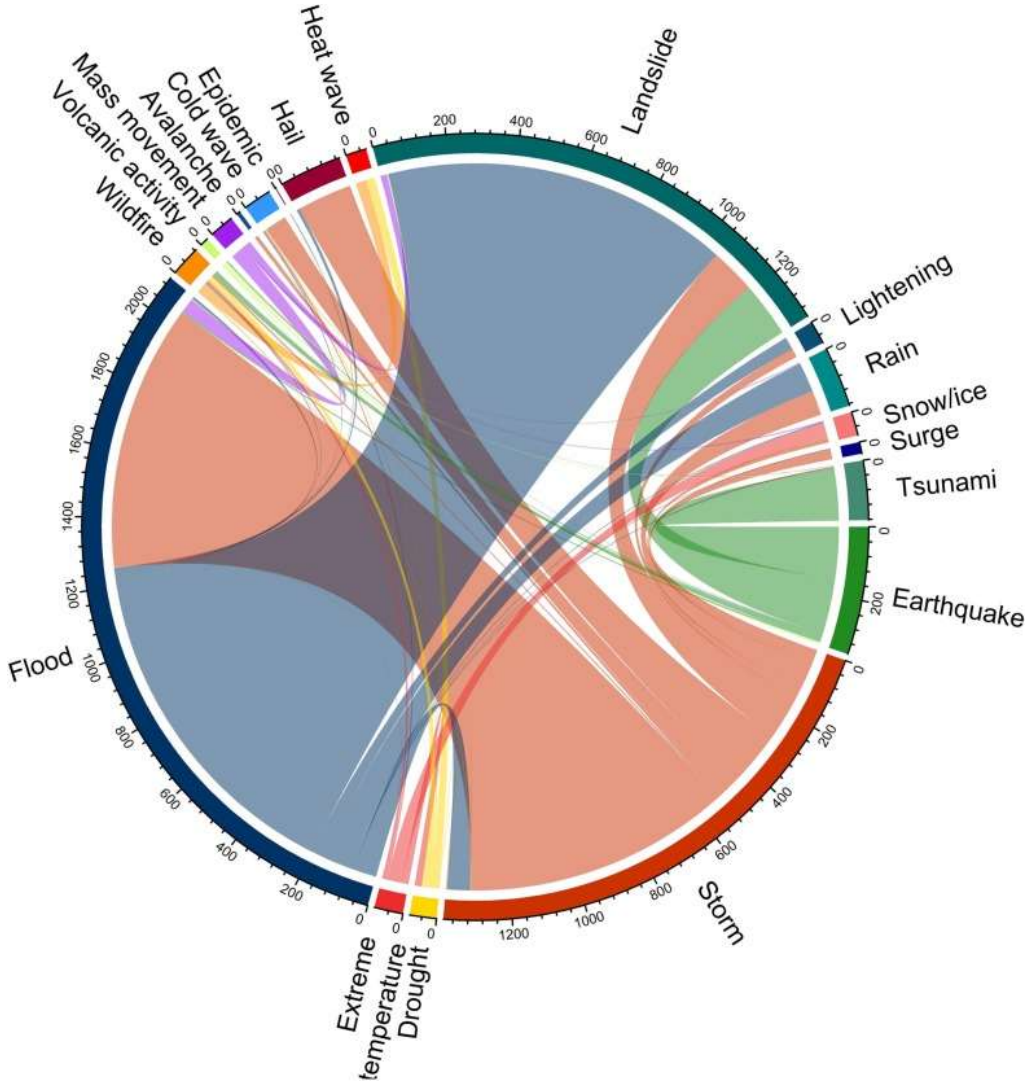


TABLE 3.2

Countries most affected by climatic disasters, 1990–2019

The United States has suffered the largest number of climatic disasters since 1990.

Country	Drought	Extreme temperature	Flood	Storm	Wildfire	Total
United States	14	20	145	444	81	704
China	31	13	246	264	6	560
India	7	39	216	110	3	375
Philippines	6	0	123	219	1	349
Bangladesh	1	22	65	108	0	196
Indonesia	3	0	170	5	11	189
Vietnam	6	0	85	90	1	182
Mexico	5	14	52	87	3	161
Australia	5	7	45	66	28	151
Japan	0	16	27	103	1	147
Russia	5	21	66	21	24	137
Brazil	10	4	95	10	4	123
France	2	18	41	56	6	123
Pakistan	2	15	84	20	0	121
Thailand	12	2	68	33	1	116
Afghanistan	5	7	85	9	1	107
Haiti	5	0	45	34	0	84
Canada	0	4	33	28	17	82
Argentina	4	7	48	16	4	79
Colombia	2	0	68	5	3	78

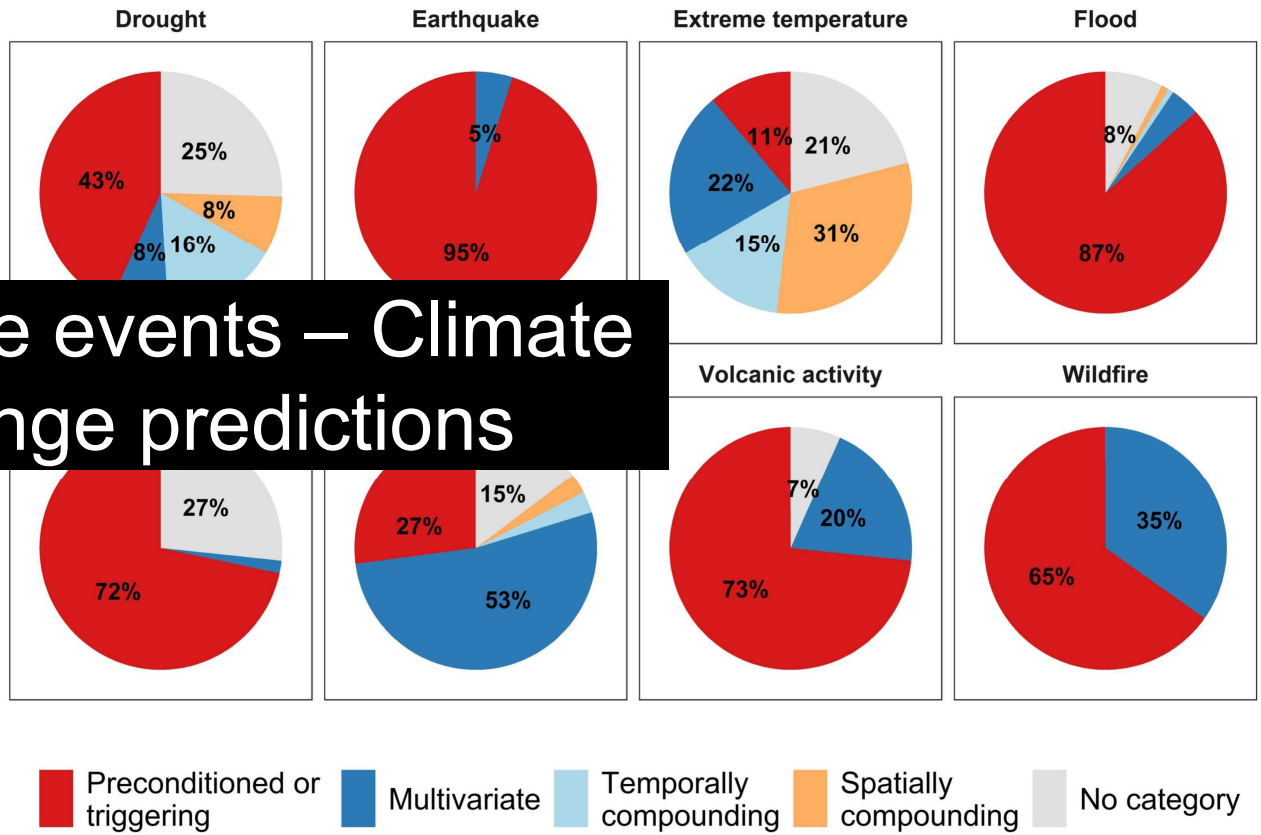
Source: EM-DAT, IEP Calculations

Lee et al. 2023 *Science of T. Env*.

Criteria for classifying multi-hazard events into four distinct categories.

Multi-hazard category	Criteria	EM-DAT example sequences(s):
Preconditioned / triggering	Prolonged driver/hazard creating antecedent conditions which exacerbate following hazard events.	*ID:* 1997–0008-USA *Year:* 1997 *Driver:* Heavy rain *Location(s):* USA *Time frame:* 38 days *Hazard:* Flood *Associated hazard:* Landslide
Multivariate	Multiple hazards, related to each other co-occurring within immediate timeframe (days).	*ID:* 2007–0556-BGD *Year:* 2007 *Driver:* Low-pressure system *Location(s):* Bangladesh
Temporally compounding	Multiple drivers/hazards occurring within a short time frame (less than 2 months), amplifying the effects of the underlying events/hazards.	*Time frame:* 30 days *Hazard:* Storm *Associated hazard:* Cold wave
Spatially compounding	Multiple 'unrelated' events within EM-DAT with identical drivers (e.g., *El Niño*) occurring at similar times, across different locations.	*ID:* 2005–0655-BEL, 2005–0655-DEU, 2005–0655-FRA *Driver: El Niño* influenced Atlantic storms *Location(s):* Belgium, Germany, France *Time frame:* 4 days *Hazard:* Extreme Temperature *Associated hazard:* Snow/Ice

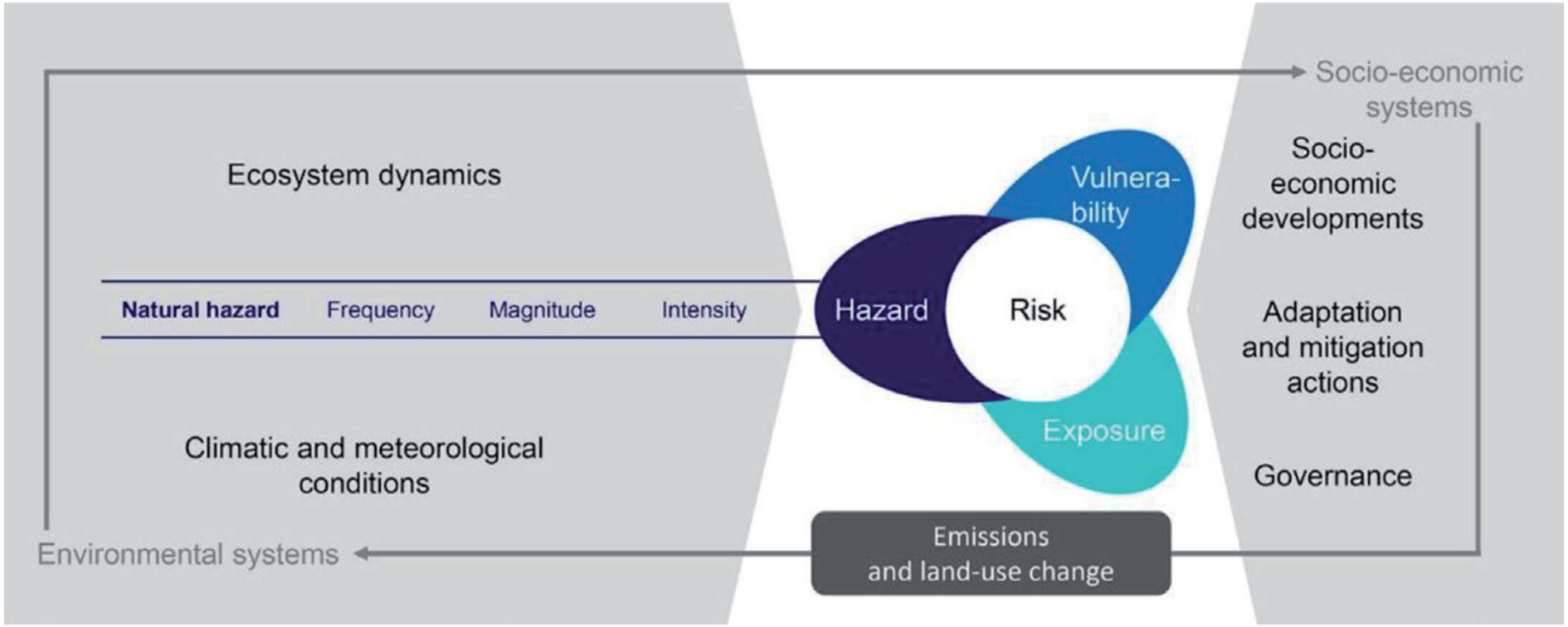
Distribution of multi-hazard events from 1900 to 2023



Extreme events – Climate change predictions

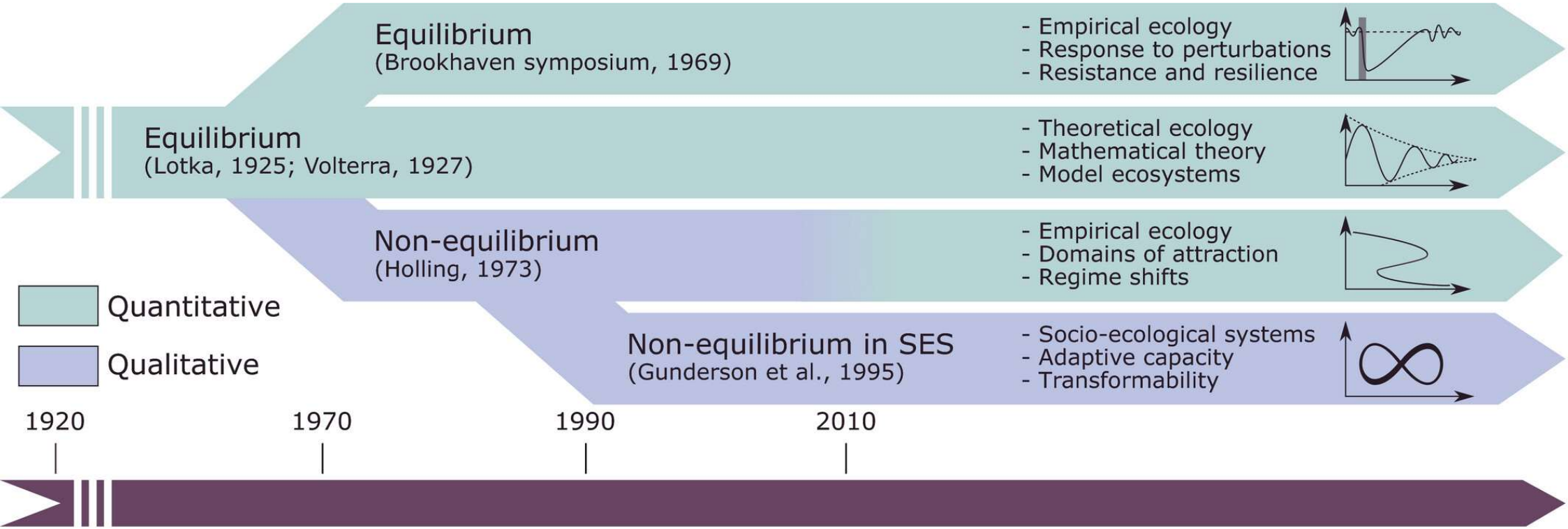
Lee et al. 2023 *Science of T. Env.*

NATURAL HAZARDS MOSTLY ASSESS ANTHROPOGENIC IMPACTS. BUT HOW ABOUT THE ENVIRONMENT EFFECTS?



IPCC 2014

CONCEPT OF ECOSYSTEM STABILITY AFTER A NATURAL HAZARD



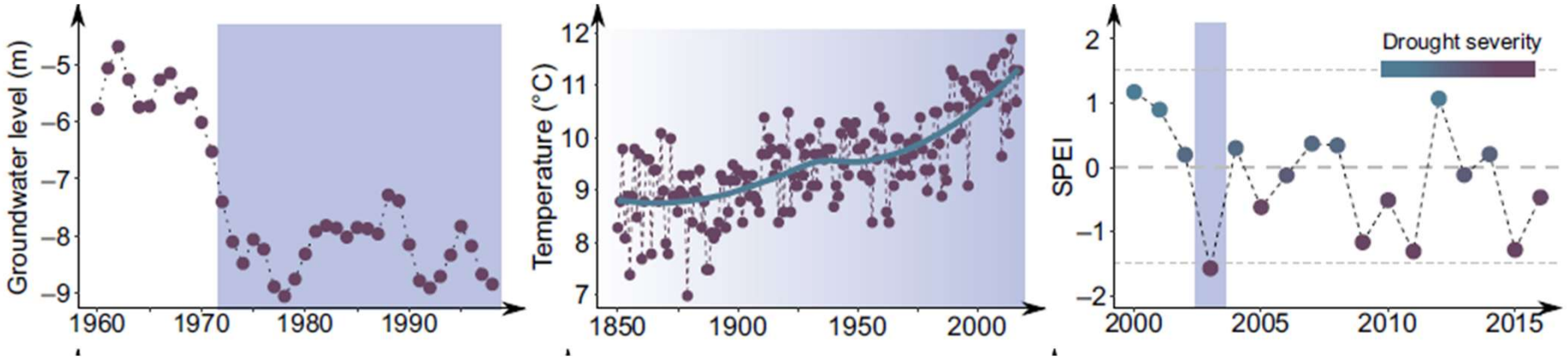
Van Meerbeek et al 2021 *Journal of Ecology*

CHARACTERISTICS OF ECOSYSTEM STABILITY

Variables *(S)* → *Biomass, vegetation cover, etc*

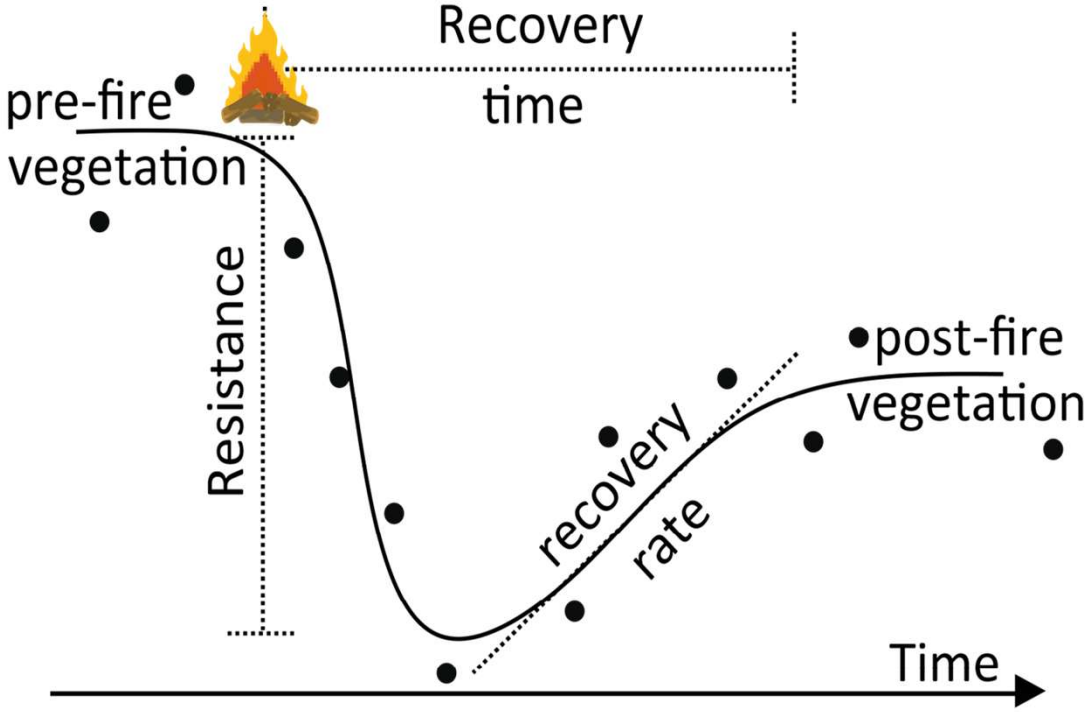
Parameters *(P)* → *factors that influence the variables of the system (e.g. temperature, precipitation, fire, hurricanes, etc)*

Stability*: it is a comparison with a reference condition of the system´s descriptors (S and P).*



STABILITY PROPERTIES: RESILIENCE, RESISTANCE, AND RECOVERY

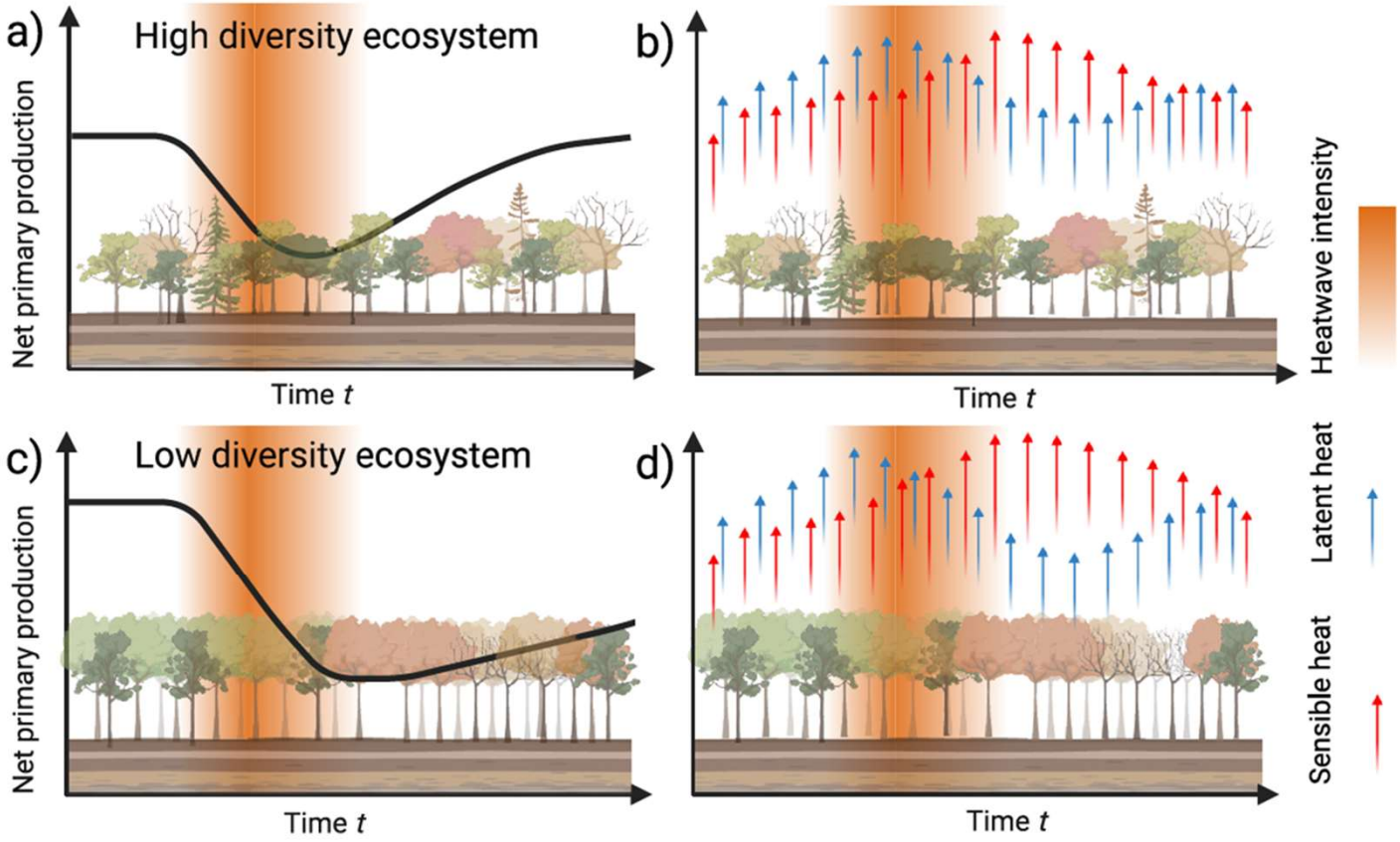
Property	Short definition	Synonyms and related terms
Resistance	The ability to withstand changes	Inertia
		Reactivity
		~Robustness
		~Sensitivity
Resilience	Rate of return after a displacement	Elasticity
		Resiliency
		Stability
		~Adjustment stability
		Engineering resilience
		Recovery
		Asymptotic stability
Recovery	Ability to fully recover	Malleability
		Resilience



Adapted from Castagneri et al. 2022 *Ecosystems*

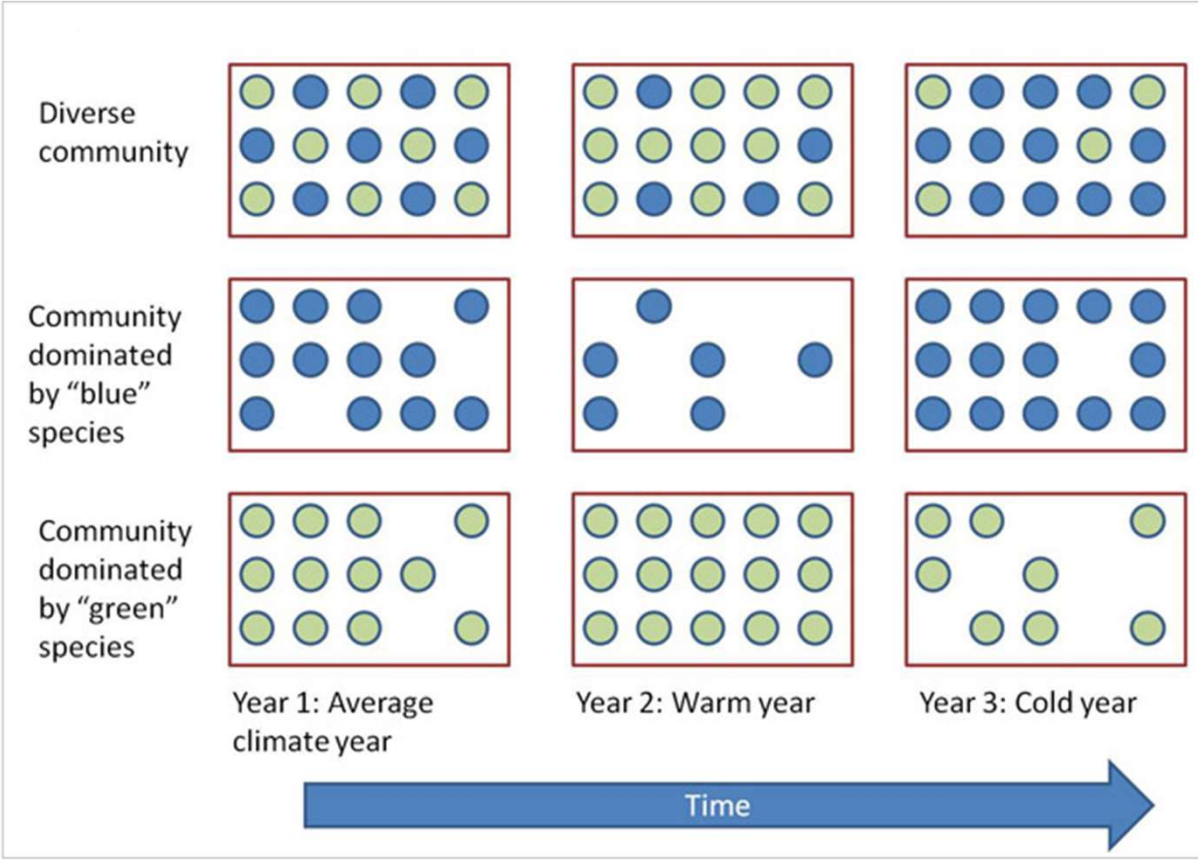
Van Meerbeek et al 2021 *Journal of Ecology;* Cole et al. 2017 *PAGES*

Insurance hypothesis - Disturbance



Mahecha et al. 2024 *Earth´s Future*

BIODIVERSITY AND ECOSYSTEM RESILIENCE



Robert May (1973):
"Increasing diversity would decrease dynamic stability"

New theories propose that diversity may not increase the dynamic stability but also not decrease it either."

Adapted from Cleland, 2011, Biodiversity and Ecosystem Stability. *Nature Ed. K.*

MEASURES OF STABILITY

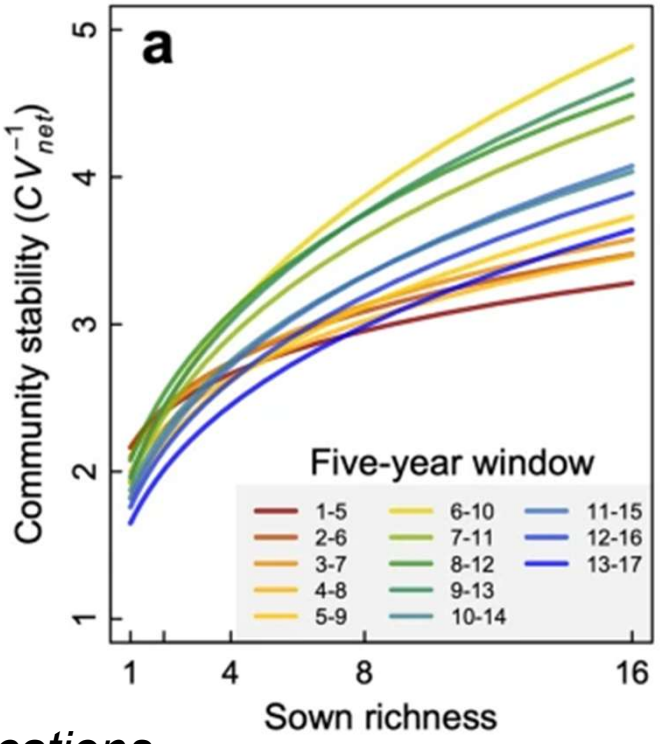
The coefficient of variation

$$CV = \frac{\text{Standard deviation}}{\text{Mean}}$$

"Community stability depends on a negative covariance effect (competitive compensation) and on overyielding (ecosystem productivity increasing with diversity."

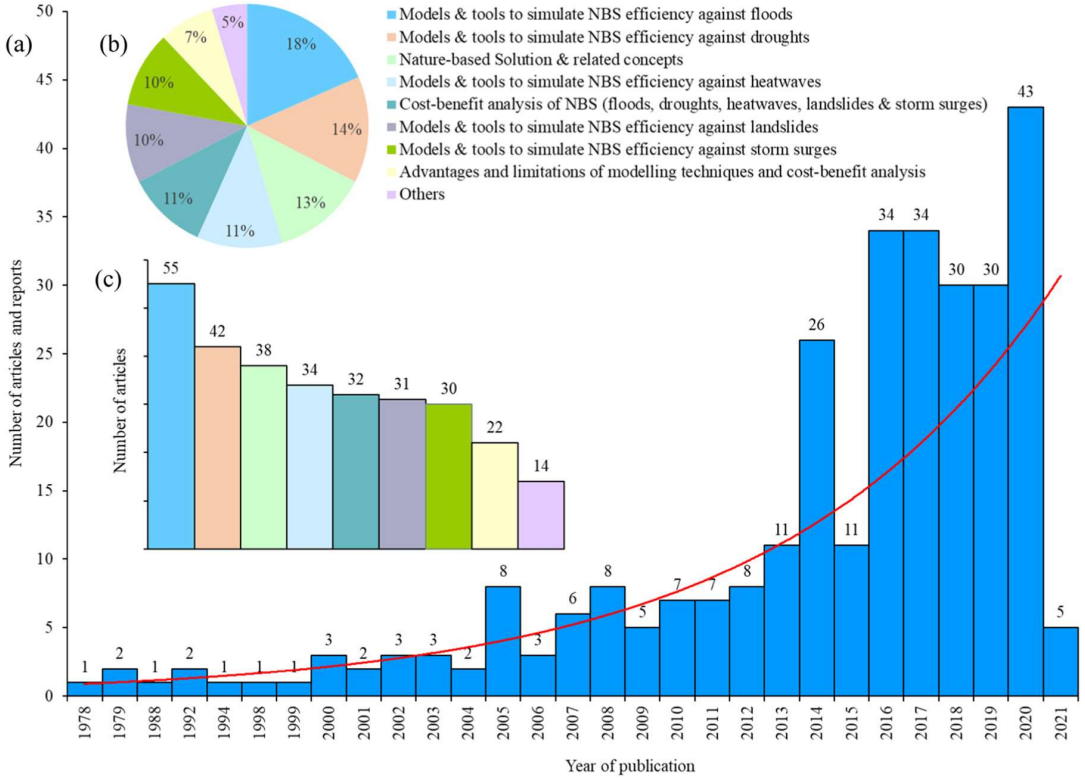
The inverse of the coefficient of variation

1/ CV

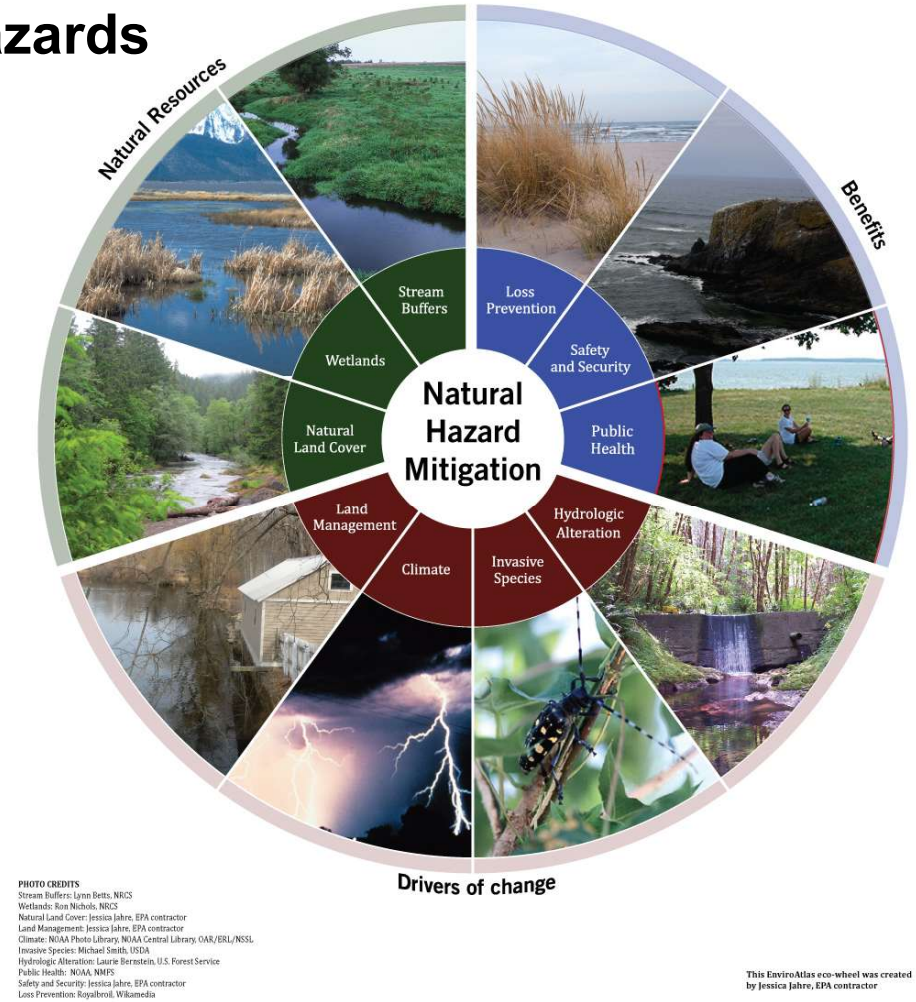


Tilman 1999, Ecology; Wagg et al. 2022 *Nature Communications*

Nature-based Solutions can mitigate natural hazards



Kumar et al. 2021b



EPA, United States Environ. Protection Agency

SUMMARY OF THE SEMINAR

- ✓ Differences between natural hazards, disasters and risks.
- ✓ Quantitative consequences of natural hazards in a biogeographical perspective
- ✓ Nature-Based solutions: natural hazard mitigation
- ✓ Ecological measures of how do ecosystem can respond to multiple natural hazards (Stability theory: resilience and resistance to a disturbance)

TODAY'S EXERCISE: GROUP DISCUSSION

In a group of 4 people:

Goal: practice diagnosing how resilience/resistance is *measured*.



15 min

1. Choose one event of a natural hazard.
2. What type of natural hazard (Geophysical, Meteorological, hydrological, biological, etc.)
3. Make notes about: Location, magnitude, occurrence and future probability.

TODAY´S EXERCISE IN GROUPS



15 min

Each group :

Present your solution:

1. In a biogeographical perspective, what is the consequence of the chosen natural hazard for nature and for human beings?
2. What kind of measurements could have been done to estimate how resilient and resistant the ecosystem is to “the chosen natural hazard”?
3. What kind of mitigation action could be applied to avoid it?

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GROUP DISCUSSION & PRESENTATION